

**XAI for
Dummies**



User Studies

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Plan for the Day

Time	Topics
1:30 PM	Introduction
	<i>Hello and Introductions!</i>
1:45 PM	From Philosophy of Counterfactuals to XAI and User Studies
	<i>Philosophical & Psychological Fundamentals of Counterfactuals</i>
	<i>Getting Into the User Study Mindset</i>
	Q&A
2:20 PM	Computational Fundamentals of Counterfactuals in AI
	<i>Formalisation</i>
	<i>Current and Future Directions</i>
	Q&A
3:00 PM	COFFEE (15:00h-15:30h)
3:30 PM	Hands-On I: Using a Counterfactual Toolbox
	<i>From a Simple Baseline to Diverse Counterfactuals</i>
	Q&A
4:10 PM	Fundamentals of XAI User Studies
	<i>A Roadmap: From Motivation to Results</i>
	Q&A
4:50 PM	Hands-On II: User Sample Study - Walkthrough and Analysis
	<i>From Raw Data to a Clean Analysis Sample</i>
	<i>Testing & Interpreting Effects of Counterfactual Explanations</i>
	Q&A
5:20 PM	Closing Session, Discussion and Final Q&A
TUTORIAL END	

You Are Here!

The Roadmap: Six Steps to Heaven...





Motivation

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AI explanations are meant for humans - we need to test if they actually work

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User studies in XAI generally:

- test the effects of explanation use

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User studies in XAI generally:

- test the effects of explanation use

The first key task:

Define a precise research question:



To **what end** do you want to test **what**
with **whom** in which **context**?



Motivation

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To what end do you want to test what with whom in which context?

Research question:

How do counterfactual explanations for
AI-generated hiring decisions impact job applicants'
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How do counterfactual explanations for
AI-generated hiring decisions impact job applicants'
trust?

Hypothesis:

We assume that counterfactual explanations improve the
trust of AI-generated hiring decisions by job
applicants, compared to black box decisions.

Make hypotheses precise and testable!

From Research Question to Design

We assume that counterfactual explanations improve the **trust** of **AI-generated hiring decisions** by **job applicants**, compared to black box decisions.



Independent Variable

What do we vary?

→ CFE vs. no explanation

From Research Question to Design

We assume that counterfactual explanations improve the **trust** of **AI-generated hiring decisions** by **job applicants**, compared to black box decisions.

Independent Variable

What do we vary?

→ CFE vs. no explanation

Dependent Variable

What do we measure?

→ trust

From Research Question to Design

We assume that counterfactual explanations improve the **trust** of **AI-generated hiring decisions** by **job applicants**, compared to black box decisions.

Two-group study is a go-to standard!



Control:
No-Explanation
(NO-XP)



Experimental:
CF-Explanation
(CF-XP)



Design

Interlude: Consider Power...

How Many People Do We Test?



Design

Let's Consider Power

How Many People Do We Test?

- Most common user-study flaw: **too few** participants
- ... but you can also test too many !

Let's Consider Power

How Many People Do We Test?

- Most common user-study flaw: **too few** participants
- ... but you can also test too many !
- **Type I Error** = false positive:
Detect an effect that is not actually there
--> might happen if N is too high!
- **Type II Error** = false negative:
Overlook an effect that is there
--> low N

Power Analysis tells you the optimal N for your design based on your estimate of effect-size for a p-level

Let's Consider Power

How Many People Do We Test?

- Most common user-study flaw: **too few** participants
- ... but you can also test too many !

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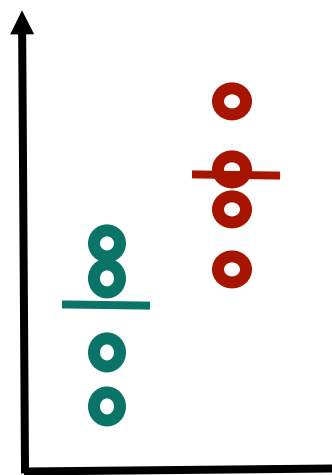
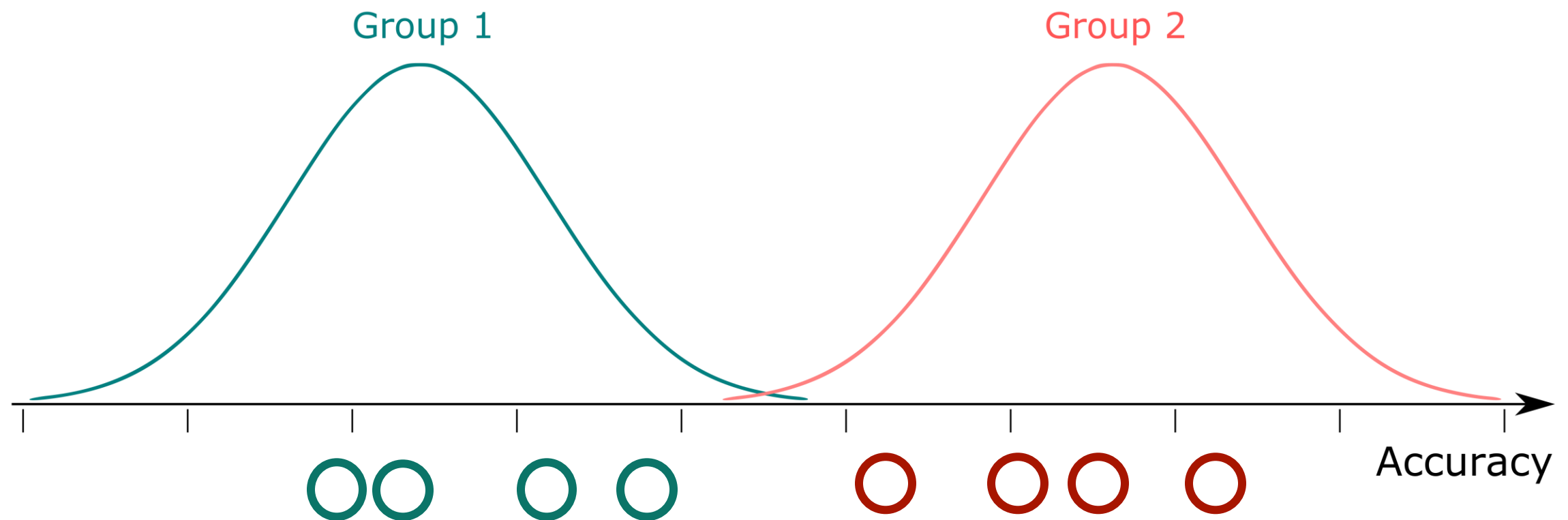
Rule of thumb: More Variables = More People!

- *Type II Error* = false negative:
Overlook an effect that is there
--> might happen if N is too high!

Power Analysis tells you the optimal N for your design based on your estimate of effect-size for a p-level

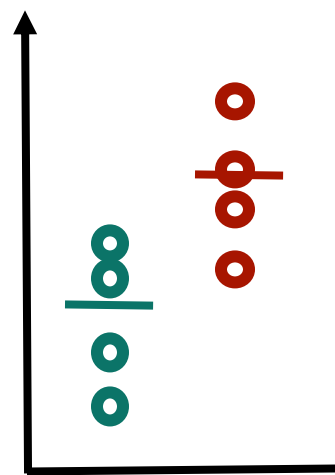
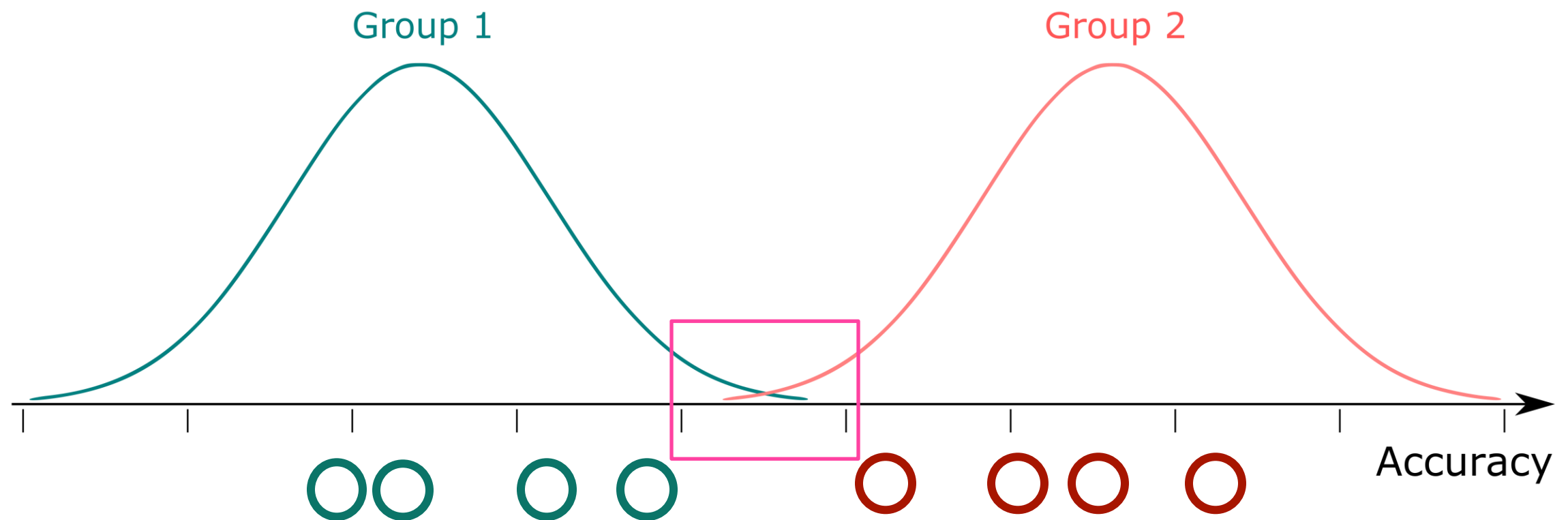


What Is Power?



--> test will produce small p-value,
correctly indicating the effect!

What Is Power?



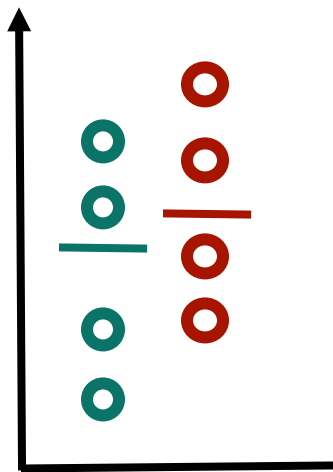
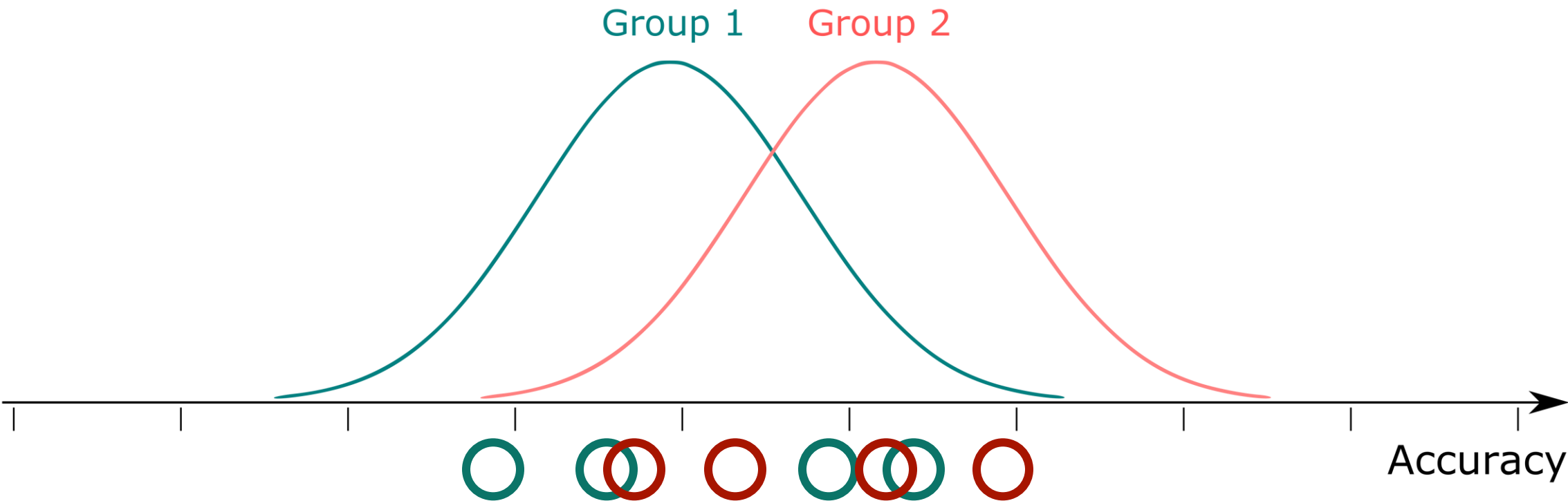
Small overlap =
high probability to correctly find the effect
(even with small samples) = high power

Power is the probability of **correctly**
finding an existing effect.

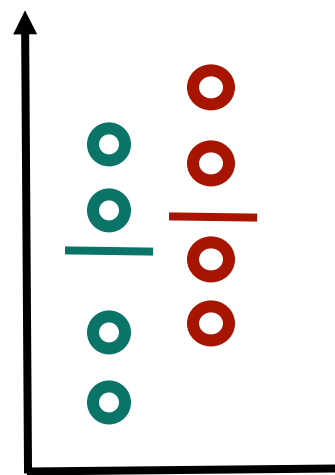
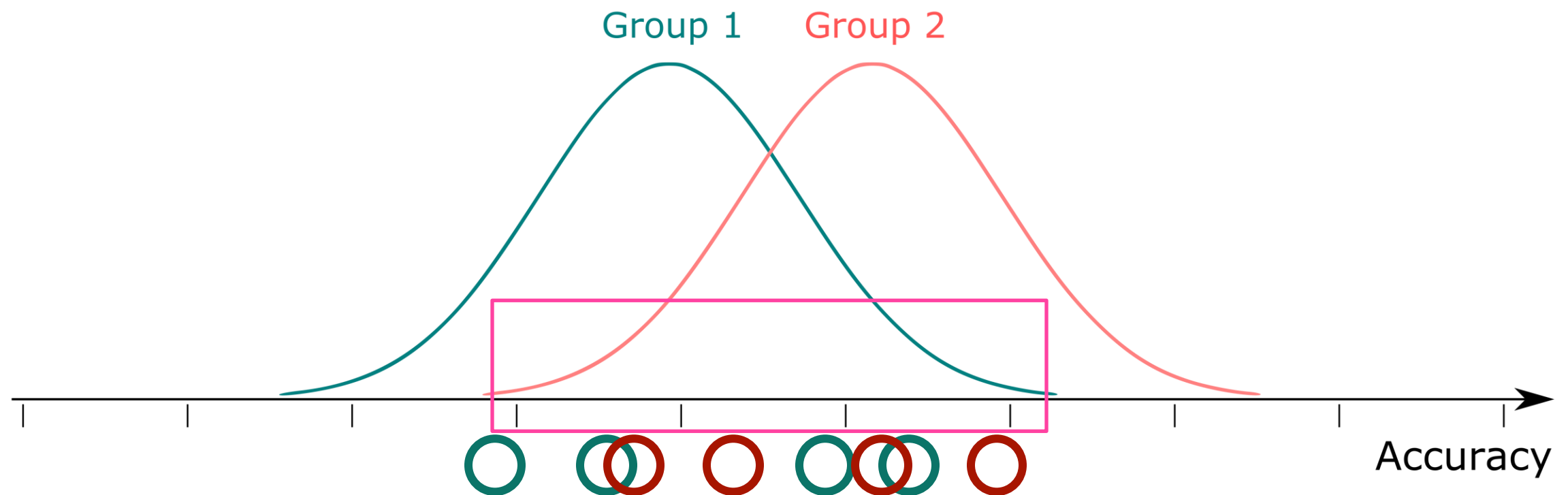


Design

What Is Power?



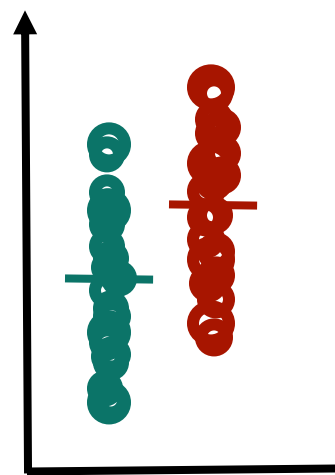
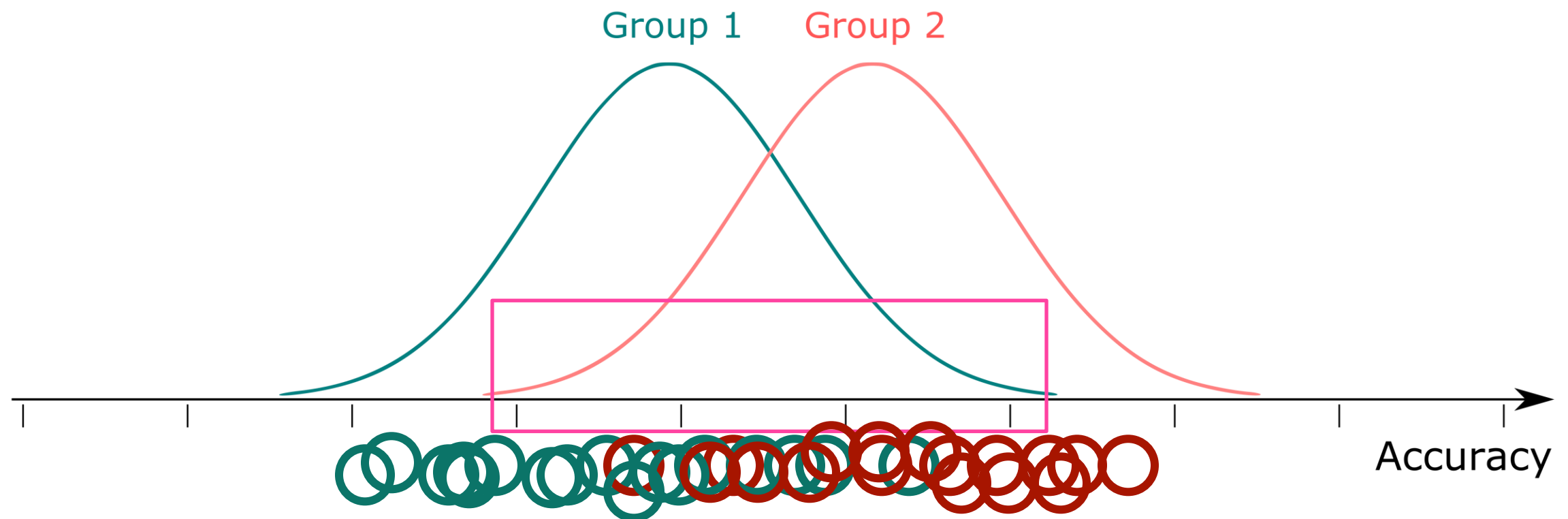
What Is Power?



Big overlap =
low probability to correctly find the effect
(esp. with small samples) = low power

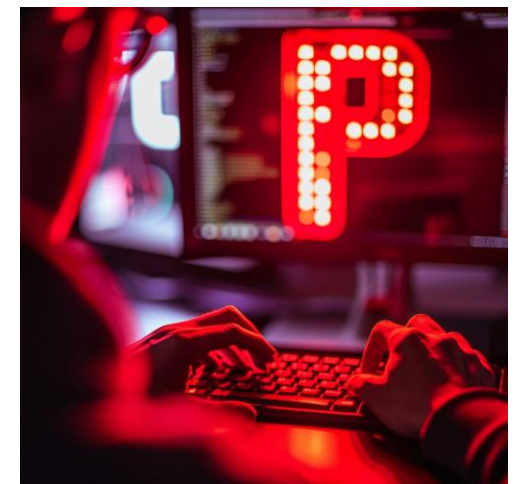
Power is the probability of **correctly**
finding an existing effect.

What Is Power?



Good news! More samples = more power!

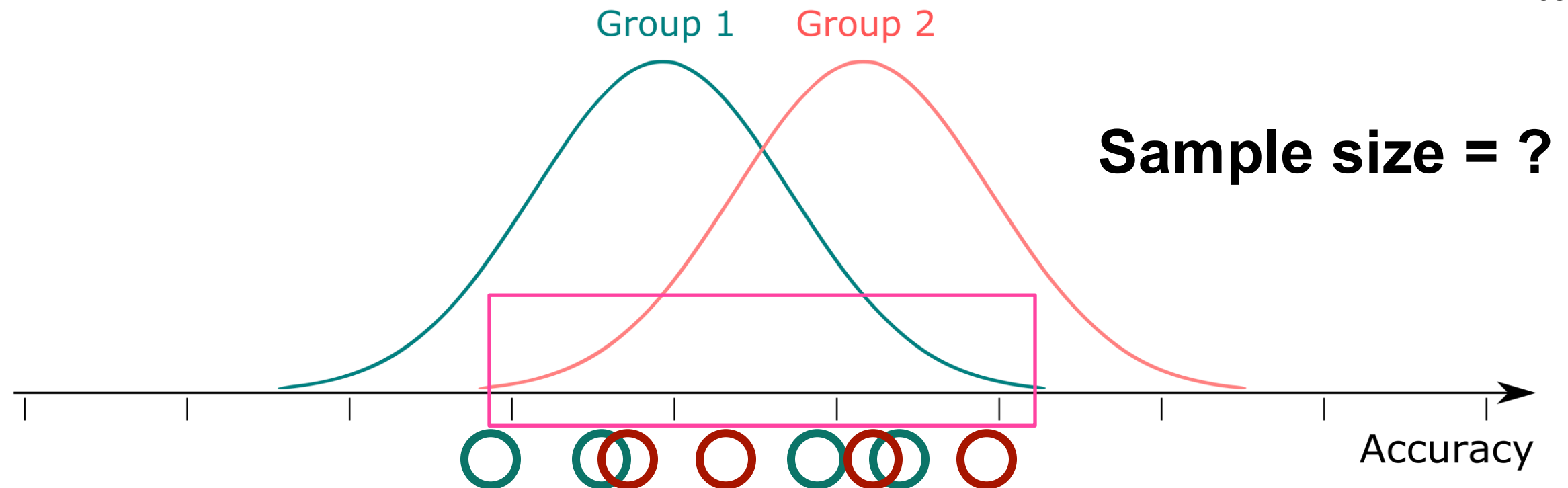
But: Do not just keep sampling until the effect is there:
p-hacking is evil!





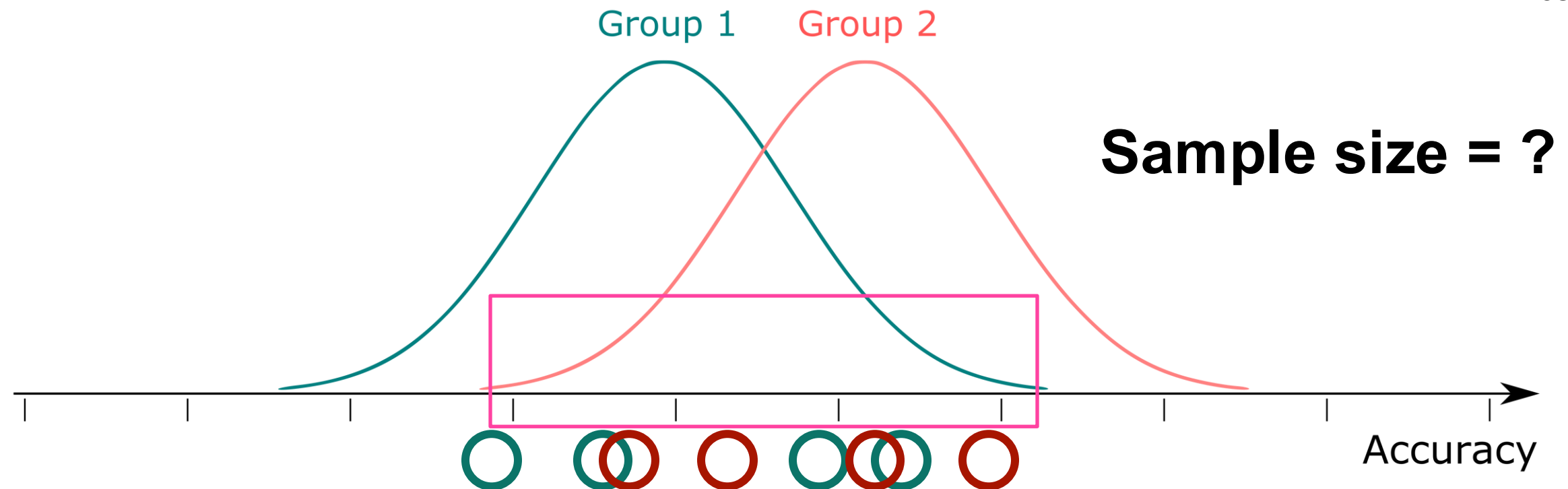
Design

Necessary Decisions



1. How much power do we want? Convention: Power = 0.8
Meaning: we want an 80% probability of correctly finding an effect.

Necessary Decisions



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Meaning: we want an 80% probability of correctly finding an effect.
2. What is our significance level α ? Convention: $\alpha = 0.05$
Meaning: „false-alarm rate“ we are willing to accept when there is actually no effect.
3. How much overlap between our distributions do we expect?
Tricky! Literature? Pilot data? Educated guess?



Design

Necessary Decisions

Single metric to combine both:

$$\text{Effect Size (d)} = \frac{\text{Estimated difference between means}}{\text{Estimated standard deviation}}$$

Sample size = ?

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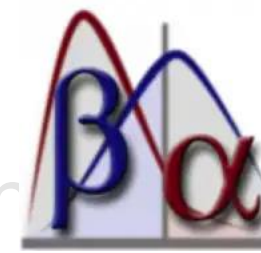
Necessary Decisions

Single metric to combine both:

$$\text{Effect Size } (d) = \frac{\text{Estimated difference between means}}{\text{Estimated standard deviation}}$$

Our case: we have

- 2 independent groups
- want power = 0.8
- $\alpha = 0.05$
- *medium effect size* $d = 0.5$



G*Power:

We need: **88**

participants per group!

3. How much overlap between our distributions do we expect?

Tricky! Literature? Pilot data? Educated guess?

Materials: Everyone Sees the Same Items



Control:
No-Explanation
(NO-XP)



Experimental:
CF-Explanation
(CF-XP)



What are we going to show people?

How To Get Those Items

*random sample
(40 from 1000s)*



data-set/domain

Candidate 378 (male)	
Experience	NA
Education	5
Soft Skills	9
Decision	Not Hired

Candidate 723 (female)	
Experience	10
Education	5
Soft Skills	9
Decision	Not Hired

Candidate 11 (male)	
Experience	5
Education	NA
Soft Skills	8
Decision	Not Hired

Candidate 412 (male)	
Experience	4
Education	4
Soft Skills	4
Decision	Not Hired

How To Get Those Items

*random sample
(40 from 1000s)*



data-set/domain

Candidate 378 (male)	
Experience	NA
Education	5
Soft Skills	9
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Candidate 723 (female)	
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Candidate 11 (male)	
Experience	5
Education	NA
Soft Skills	8
Decision	Not Hired

Candidate 412 (male)	
Experience	4
Education	4
Soft Skills	4
Decision	Not Hired

How To Get Those Items

constrained sample

- with all features of interest*
- for counterfactuals: near decision boundary*
- balanced positive / negative outcomes ...and so on*



data-set/domain



material-set

Candidate 312 (male)	
Experience	8
Education	7
Soft Skills	8
Decision	Hired

Candidate 965 (female)	
Experience	8
Education	5
Soft Skills	3
Decision	Not Hired

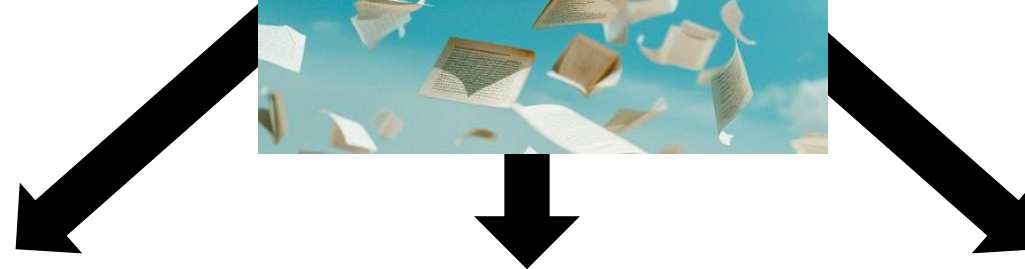
Candidate 681 (male)	
Experience	4
Education	5
Soft Skills	6
Decision	Not Hired

Candidate 34 (female)	
Experience	3
Education	8
Soft Skills	9
Decision	Hired

Is Material-Set Balanced?



Materials &
Procedure



Candidate M1	Candidate M2	Candidate M3	Candidate M4	Candidate M5	Candidate M6	Candidate M7	Candidate M8

$\frac{1}{2}$ items are positive $\frac{1}{2}$ items are negative

Is Material-Set Balanced?



Materials &
Procedure



Candidate M1	Candidate M2	Candidate M3	Candidate M4	Candidate M5	Candidate M6	Candidate M7	Candidate M8
male	male	female	female	female	male	female	male

$\frac{1}{2}$ items are positive $\frac{1}{2}$ items are negative

$\frac{1}{2}$ items have male gendered, $\frac{1}{2}$ items have female gendered

Is Material-Set Balanced?



Materials &
Procedure



Candidate M1	Candidate M2	Candidate M3	Candidate M4	Candidate M5	Candidate M6	Candidate M7	Candidate M8
male	male	female	female	female	male	female	male
Soft Skills	Education	Soft Skills	Education	Soft Skills	Soft Skills	Education	Education

$\frac{1}{2}$ items are positive $\frac{1}{2}$ items are negative

$\frac{1}{2}$ items have male gendered, $\frac{1}{2}$ items have female gendered

each item based on one of two key features

(here: Soft Skills, Education), balanced between groups / gender

Is Material-Set Balanced?



Materials &
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Candidate M1	Candidate M2	Candidate M3	Candidate M4	Candidate M5	Candidate M6	Candidate M7	Candidate M8
male	male	female	female	female	male	female	male
Soft Skills	Education	Soft Skills	Education	Soft Skills	Soft Skills	Education	Education

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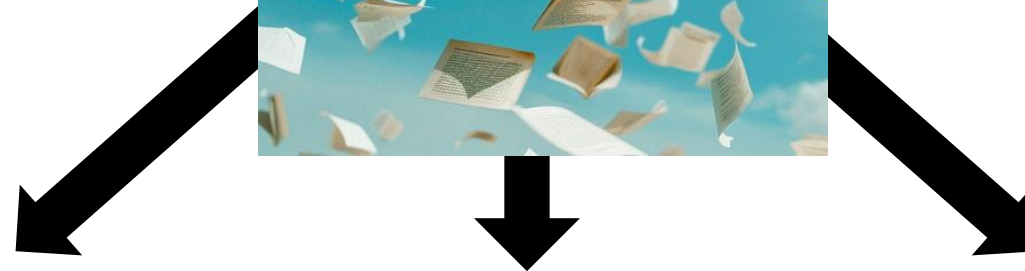
(here: Soft Skills, Education), balanced between groups / gender

*The ultimate challenge: Is the material set balanced for **hidden** variables? Balancing can be hard (impossible)!*

Is Material-Set Balanced?



Materials &
Procedure



Candidate M1	Candidate M2	Candidate M3	Candidate M4	Candidate M5	Candidate M6	Candidate M7	Candidate M8



Candidate M2	Candidate M3	Candidate M7	Candidate M1	Candidate M8	Candidate M5	Candidate M6	Candidate M4



Candidate M5	Candidate M1	Candidate M6	Candidate M8	Candidate M2	Candidate M3	Candidate M7	Candidate M4

Avoid order effects: randomly re-order items for each person !

Procedure: Task

What will users do?



Materials &
Procedure



Control:
No-Explanation
(NO-XP)



Experimental:
CF-Explanation
(CF-XP)

Candidate M2 (male)	
Experience	4
Education	5
Soft Skills	6
Decision	

Participants: Read the item

- evaluate case
- get No/CF-XP
- give trust rating
- next item

Procedure: Task

What will users do?



Materials &
Procedure

NO-XP

“The system is trustworthy.”

Strongly Disagree	Disagree	Undecided	Agree	Strongly Agree
(1)	(2)	(3)	(4)	(5)

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Materials & Procedure

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CF-XP

If M2 had Education + 2, he would be hired.

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Procedure: Task

What will users do?



Materials & Procedure

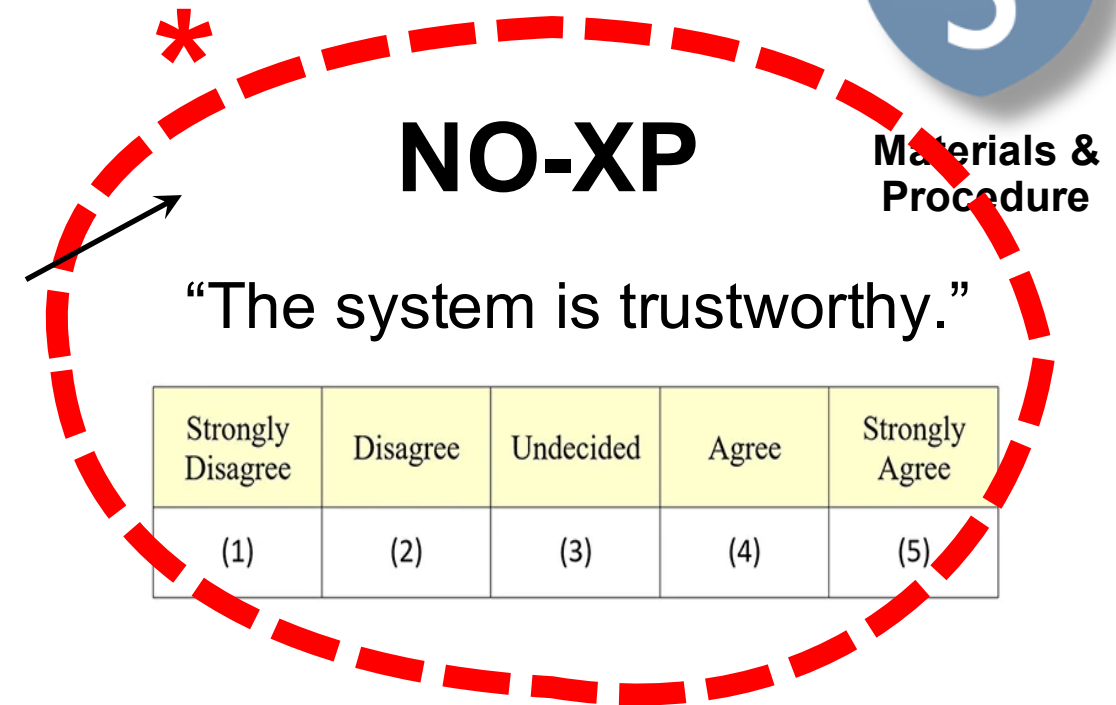


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CF-XP

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Participants: Read the item
 → evaluate case
 → get No/CF-XP
 → give trust rating
 → next item

*** Can't Be the Same? But Need to be Matched!**
 → Cognitive Load!

Procedure: Task

What will users do?



Materials & Procedure

NO-XP

M2 was hired because of his features.

“The system is trustworthy.”

Strongly Disagree	Disagree	Undecided	Agree	Strongly Agree
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CF-XP

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Participants: Read the item
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→ get No/CF-XP
→ give trust rating
→ next item

Bottom line: keep it very simple with clear instructions

Procedure: The Hidden Stuff

That Can Go Un-noticed

- **Active vs. passive:** Asking people to do an active task (e.g. produce a prediction) → engagement!
- Task performance is an ***objective measure***
- Take care with ***subjective measures:***
 - reliable judgements possible?
 - correct vs. reasonable?

The Non-Hidden Stuff

Boilerplate: Consent, Permissions & Contact Information

- **Informed Consent**

- ~ Clear detail of study aim & procedure
- ~ Voluntariness, right to withdraw
- ~ Reimbursement & contact person



- **Privacy and Confidentiality**

- ~ Statement of confidentiality
- ~ Avoid personal data
- ~ Properly anonymized

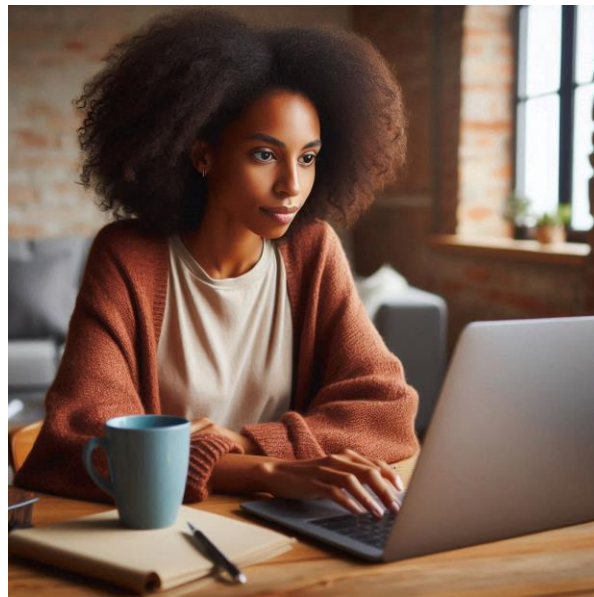


Procedure: Data Acquisition



**Materials &
Procedure**

Online



Laboratory



Act natural!

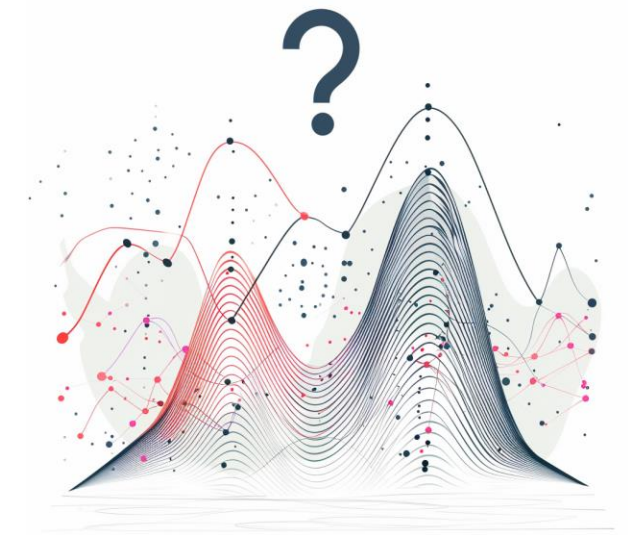


Online will tend to be cruder...

Pilot: Why Do Them?

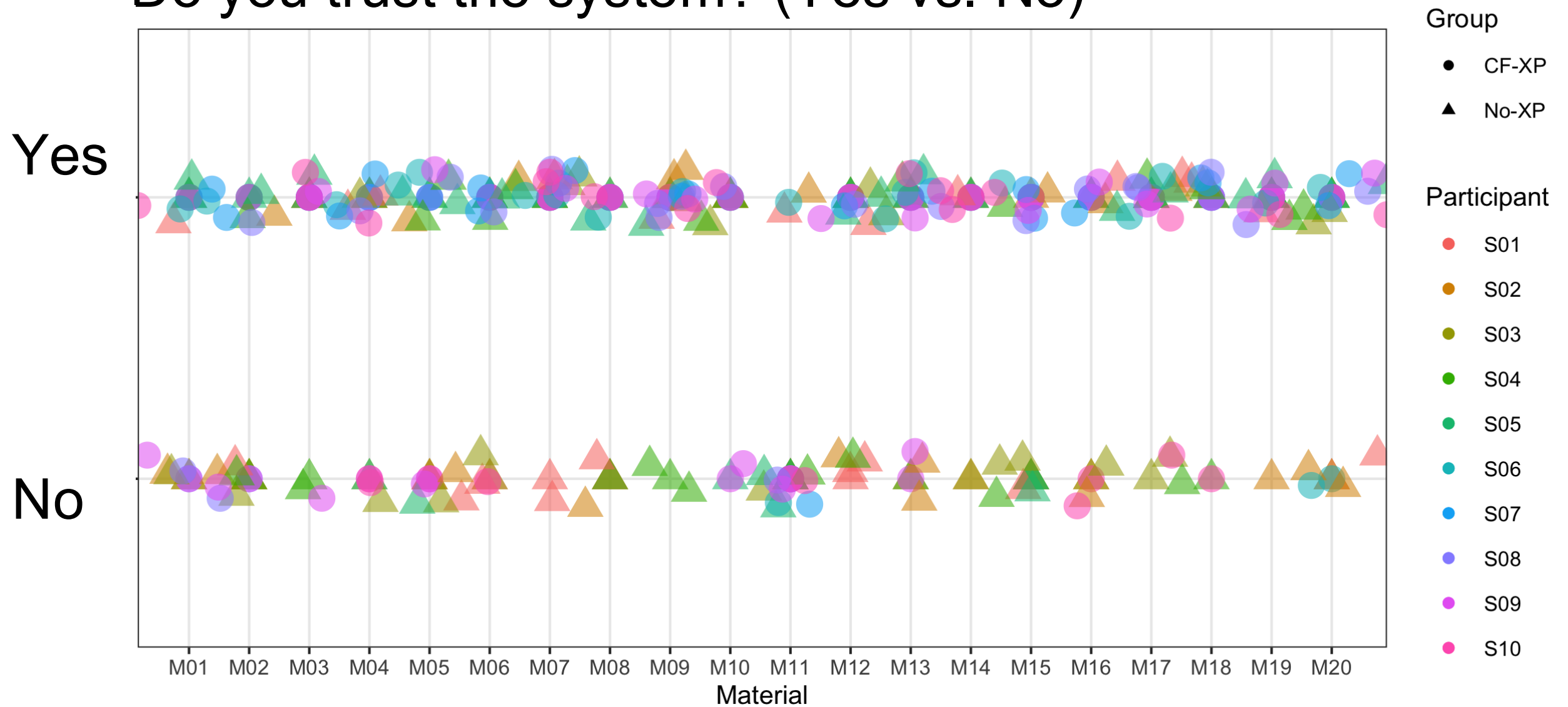
Just a Rough Test

- Pilot is run it on a small number; 5-10 people
- Issues addressed by doing pilots:
 - instructions and comprehension clear?
 - timing and fatigue
 - technical problems
 - unexpected participant strategies
- Often good to do this face-to-face (in part)
--> see people's actual reactions
- Note, stats will be meaningless but still check
→ **A pilot is for debugging the study, not testing the hypothesis!**



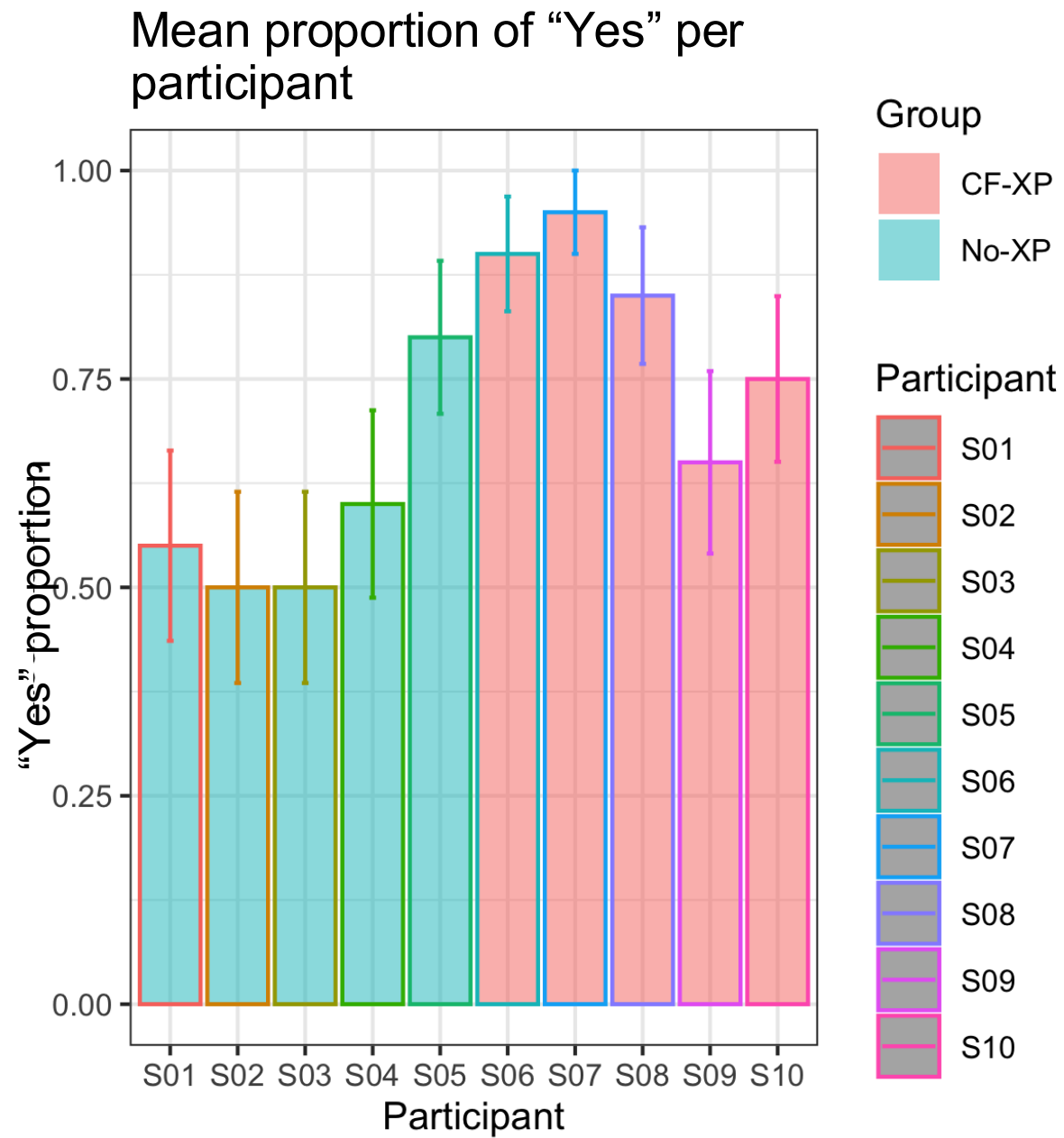
After Collecting, Cleaning !

Do you trust the system? (Yes vs. No)



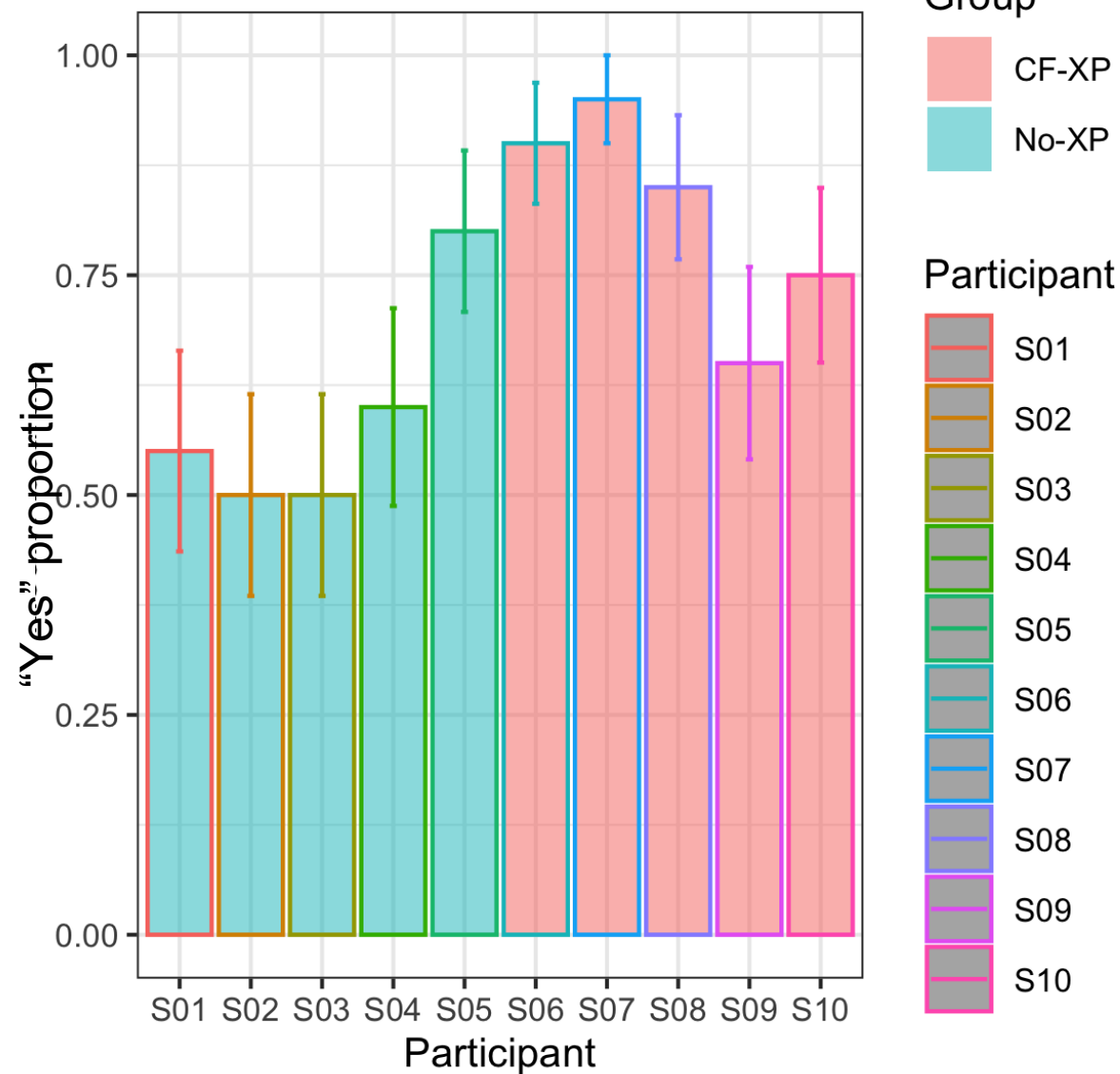
Legitimate removal of bad data-points/people...



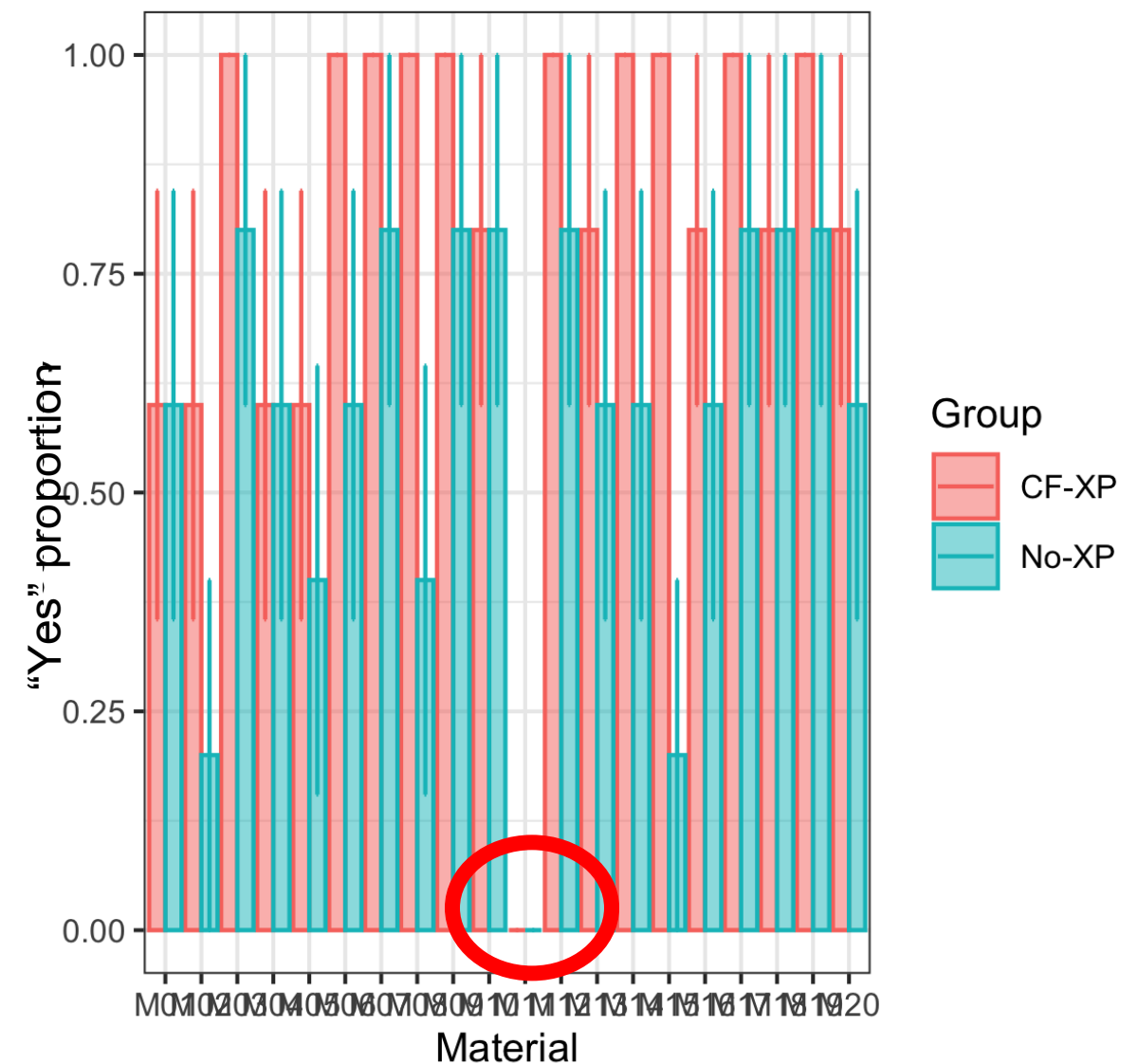


so far, so good...

Mean proportion of “Yes” per participant



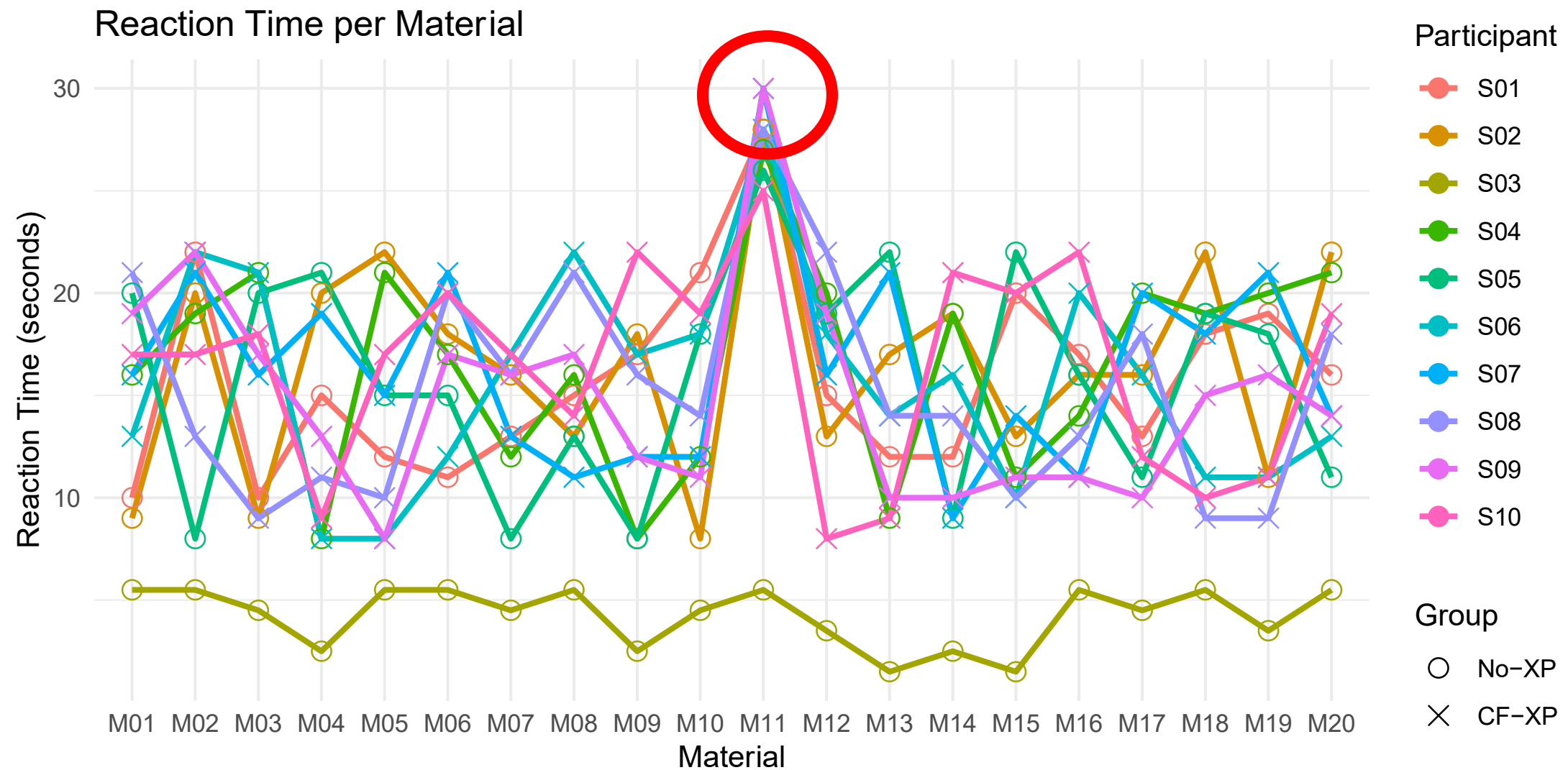
Mean proportion of “Yes” per material



so far, so good...

oops, bad material !
(pilot should normally catch this ...)

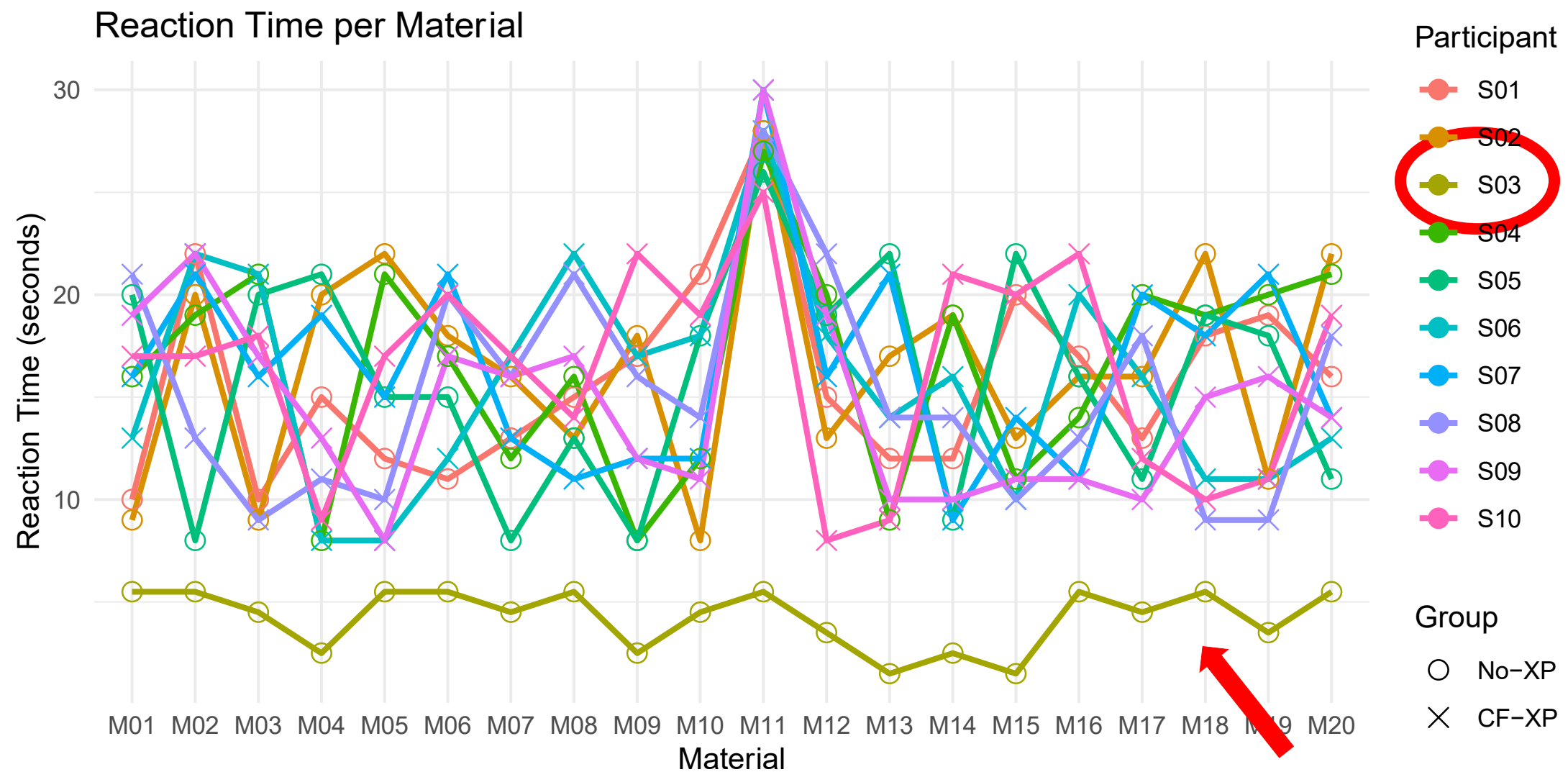
Data Collection & Analysis



Everyone is taking much longer on M11 !



Data Collection & Analysis



S03 is speeding through materials;
may also be straight-line responding !



A-priori Exclusion Criteria

Simple eyeballing the data often works well...

... but, *exclusion criteria* need to be “objective” not “selective”, ideally defined in advance !

- ~ all speedsters >3 SDs from mean
- ~ all materials >3 SDs from mean
- ~ materials >3 SDs from person's own mean, to catch attention failures
- ~ straight-liners, repeatedly giving same answer, even with negative feedback
- ~ non-varying responses



Doing the Descriptive Statistics



Mean, median, mode, standard deviation.

Make sure to describe your sample!

	Before quality assurance measures ($N = 45$)				After quality assurance measures ($N = 39$)			
	No-XP	CF-XP	U -value ^a	p -value	No-XP	CF-XP	U -value ^a	p -value
N	5	5	..	..	4	5	..	..
Gender ^b	2f/3m	3f/2m	255.5	0.950	2f/2m	3f/2m	182.5	0.788
Age (Mdn) ^c	35–44 y	35–44 y	225.5	0.516	35–44 y	35–44y	143	0.168

^aNon-parametric Wilcoxon-Mann-Whitney U -test.

^bf, female; m, male.

^c Mdn , median age band (options: 18–24 y, 25–34 y, 35–44 y, 45–54 y, 55–64 y, 65 y, and over).

Doing the Inferential Statistics



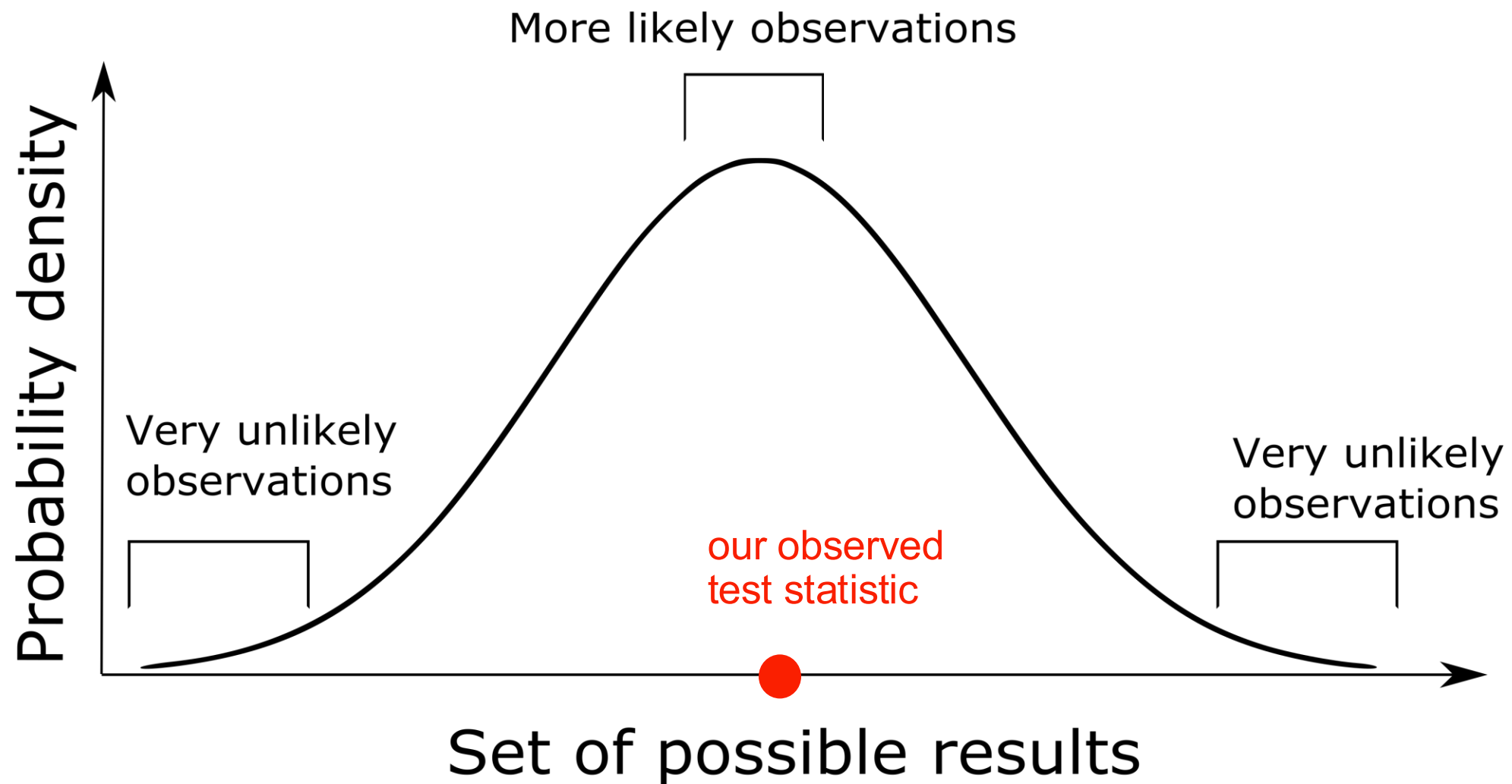
Data Collection
& Analysis



First, some basics...

General logic of statistical testing

Assuming there is no effect (null hypothesis):

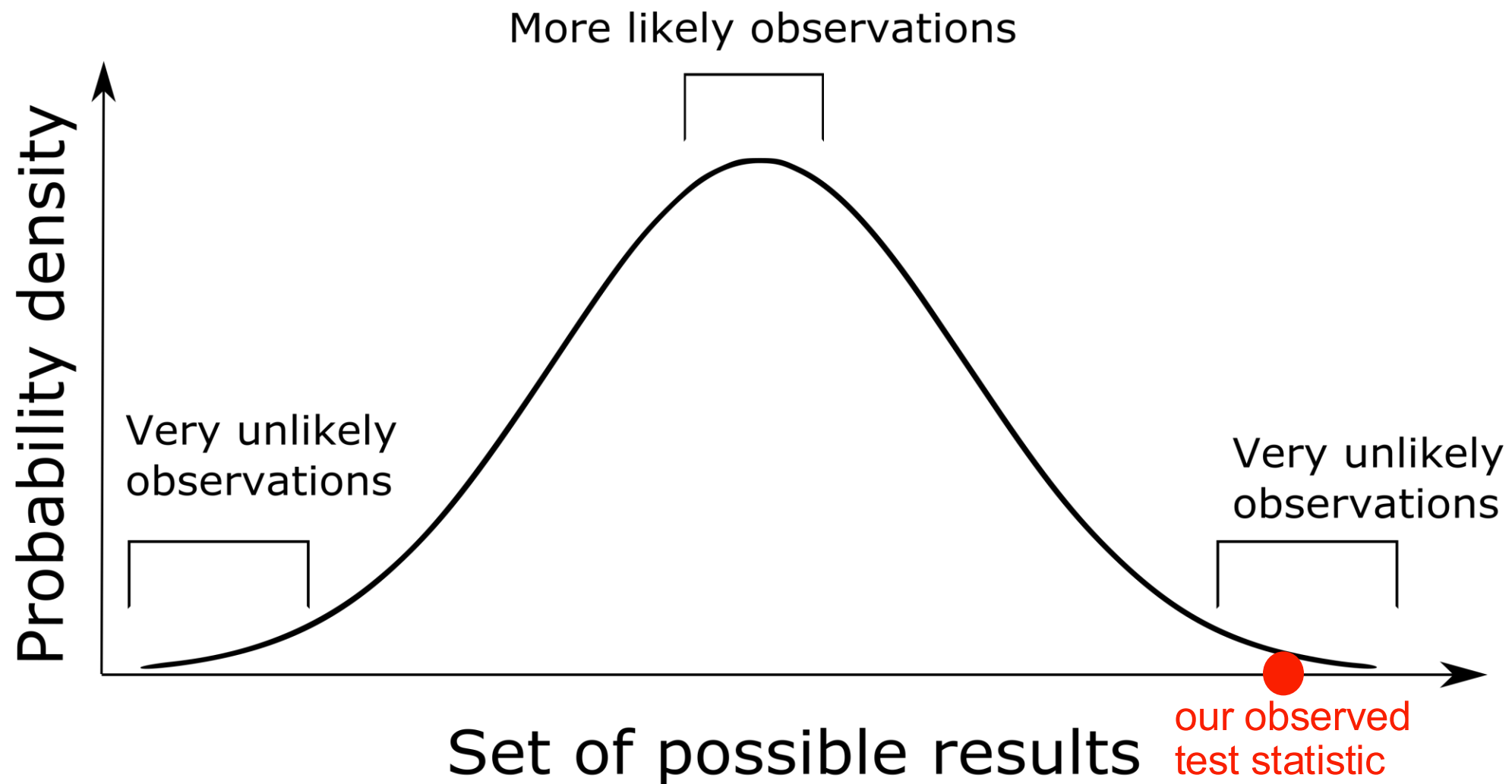


Statistical tests produce:

- a **test statistic**

General logic of statistical testing

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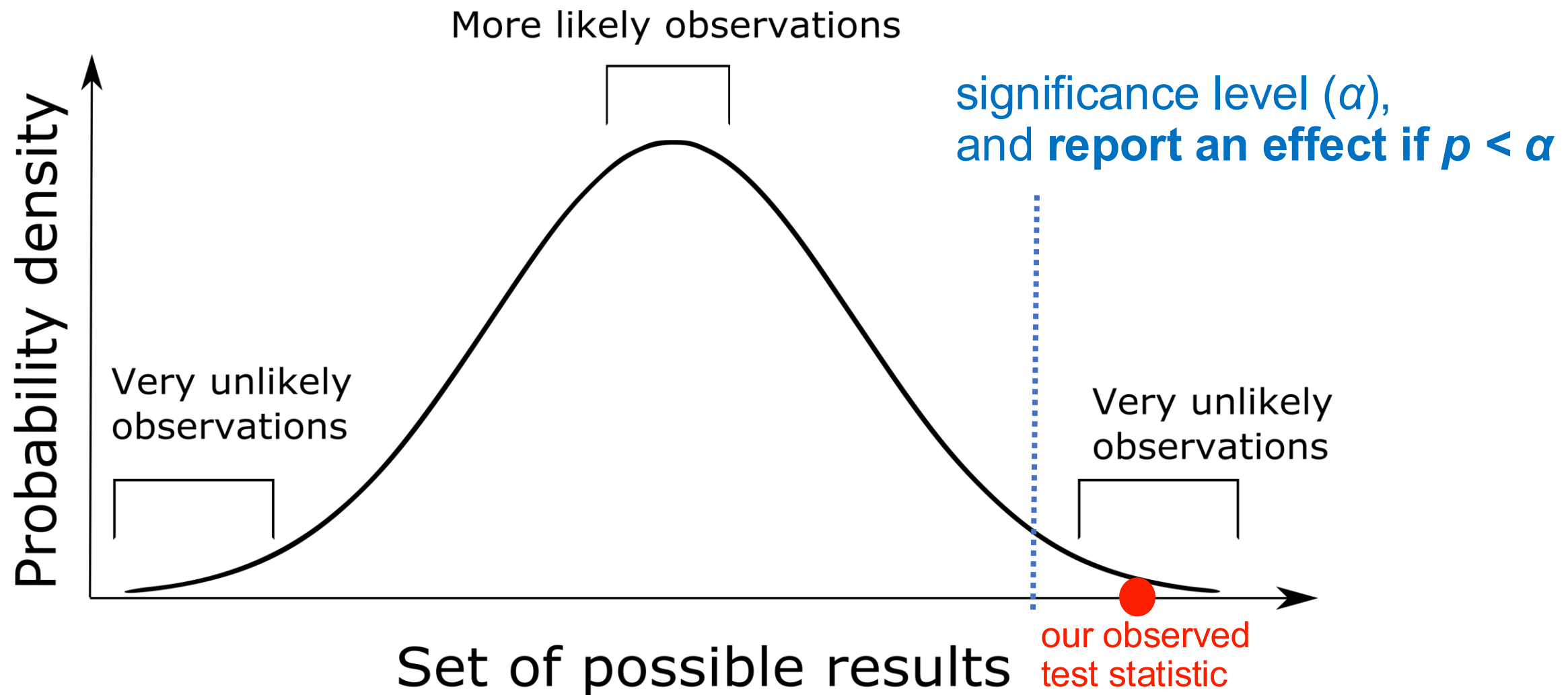


Statistical tests produce:

- a **test statistic**
- a **p-value** = likelihood that the test statistic is observed assuming no effect

General logic of statistical testing

Assuming there is no effect (null hypothesis):



Statistical tests produce:

- a **test statistic**
- a **p-value** = likelihood that the test statistic is observed assuming no effect

Doing Data Analysis: Running Inferential Statistics



Data Collection
& Analysis



Control:
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Experimental:
CF-Explanation
(CF-XP)

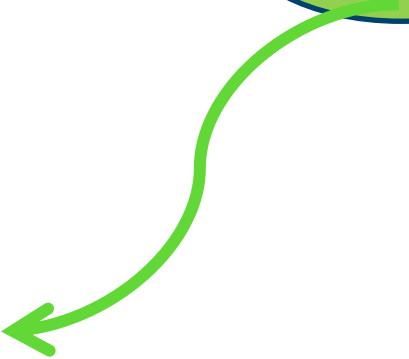
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hands-on part!



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Analyzing User Data:

https://colab.research.google.com/github/andreatelt/ICPR26_CF_Tutorial/blob/main/1-user_study_evaluation_ICPR.ipynb